

# V3 moneyness performance

Seven models · four companies · 4 windows · 1.5-year monthly calibration · classified by call moneyness  $m = S/K$

---

## What this report is

- Companion to V4\_1p5y\_monthly\_empirical\_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Ranking column ranks 1 = best. Empty buckets ( $n = 0$ ) show em dashes and have no BEST.

## Unchanged experimental design

- Same seven models, four companies (SPY, AAPL, MSFT, AMZN), 4 windows, 1.5-year lookback, monthly recalibration, V3 filters, Euler-corrected Heston step, LSM  $n_{\text{paths}} = 10000$  / seed 42.
- Same option contracts as the main study (nearest-ATM listed calls, DTE 7–60,  $|S/K - 1| \leq 10\%$ ). No model-specific re-sampling.
- Pooled  $n = 802$  contracts per model (identical keys across models).

## Moneyness rule (calls, $m = S / K$ )

- DOTM:  $m < 0.98$ . OTM:  $0.98 \leq m < 0.995$ . ATM:  $0.995 \leq m \leq 1.005$ . ITM:  $1.005 < m \leq 1.02$ . DITM:  $m > 1.02$ .
- These five bins partition the main-study filter  $|m - 1| \leq 10\%$ . The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008–2009.

## Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})^2)}$ . Ranking uses this only.
- $\text{MAE\%} = 100 \times \text{mean}(|C_{\text{model}} - C_{\text{mkt}}|/C_{\text{mkt}})$ .  $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$ .

## Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model  $\times$  Moneyness RMSE% heatmap.
- Analysis 2 — Window  $\times$  moneyness: keep the 4 windows separate, pool the four companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each window.

Table M0 · Contract counts (one model = all models)

| Regime               | DOTM | OTM | ATM | ITM | DITM | Total |
|----------------------|------|-----|-----|-----|------|-------|
| 2008-2009 Crisis     | 18   | 43  | 80  | 43  | 16   | 200   |
| 2013-2014 Normal     | 0    | 5   | 192 | 3   | 0    | 200   |
| 2018-2019 Late-cycle | 0    | 4   | 191 | 7   | 0    | 202   |
| 2019-2020 COVID      | 0    | 6   | 189 | 5   | 0    | 200   |
| All regimes          | 18   | 58  | 652 | 58  | 16   | 802   |

Moneyness rule · American calls,  $m = S/K$

| Bucket | Rule                      | n (pooled) | Share |
|--------|---------------------------|------------|-------|
| DOTM   | $m < 0.98$                | 18         | 2.2%  |
| OTM    | $0.98 \leq m < 0.995$     | 58         | 7.2%  |
| ATM    | $0.995 \leq m \leq 1.005$ | 652        | 81.3% |
| ITM    | $1.005 < m \leq 1.02$     | 58         | 7.2%  |
| DITM   | $m > 1.02$                | 16         | 2.0%  |

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

## Page 3 · Analysis 1 · Overall moneyness · Tables M1-M3

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

**Table M1 · DOTM ·  $m < 0.98$  ·  $n = 18$  · BEST = Heston**

| Model         | n         | RMSE%        | MAE%         | Bias%         | Ranking  |
|---------------|-----------|--------------|--------------|---------------|----------|
| GBM           | 18        | 50.92        | 39.60        | +27.94        | 7        |
| MD-GBM        | 18        | 35.99        | 27.39        | +3.50         | 4        |
| GARCH         | 18        | 32.41        | 25.65        | +19.46        | 2        |
| <b>Heston</b> | <b>18</b> | <b>30.32</b> | <b>25.07</b> | <b>+22.85</b> | <b>1</b> |
| Merton        | 18        | 48.70        | 37.01        | +35.27        | 6        |
| GARCH-Merton  | 18        | 43.04        | 36.75        | +35.01        | 5        |
| Heston-Merton | 18        | 33.39        | 28.41        | +25.69        | 3        |

**Table M2 · OTM ·  $0.98 \leq m < 0.995$  ·  $n = 58$  · BEST = Heston**

| Model         | n         | RMSE%        | MAE%         | Bias%         | Ranking  |
|---------------|-----------|--------------|--------------|---------------|----------|
| GBM           | 58        | 63.38        | 49.21        | +34.97        | 7        |
| MD-GBM        | 58        | 47.88        | 36.82        | +13.34        | 3        |
| GARCH         | 58        | 52.18        | 39.54        | +30.61        | 4        |
| <b>Heston</b> | <b>58</b> | <b>35.39</b> | <b>30.25</b> | <b>+21.68</b> | <b>1</b> |
| Merton        | 58        | 61.71        | 48.63        | +41.46        | 5        |
| GARCH-Merton  | 58        | 61.97        | 47.83        | +43.13        | 6        |
| Heston-Merton | 58        | 40.28        | 34.19        | +27.36        | 2        |

**Table M3 · ATM ·  $0.995 \leq m \leq 1.005$  ·  $n = 652$  · BEST = MD-GBM**

| Model         | n          | RMSE%        | MAE%         | Bias%         | Ranking  |
|---------------|------------|--------------|--------------|---------------|----------|
| GBM           | 652        | 47.99        | 38.87        | +30.16        | 4        |
| <b>MD-GBM</b> | <b>652</b> | <b>33.54</b> | <b>27.09</b> | <b>+10.33</b> | <b>1</b> |
| GARCH         | 652        | 47.02        | 35.44        | +29.21        | 3        |
| Heston        | 652        | 38.14        | 30.23        | +24.75        | 2        |
| Merton        | 652        | 50.54        | 41.37        | +35.42        | 6        |
| GARCH-Merton  | 652        | 57.61        | 45.50        | +42.06        | 7        |
| Heston-Merton | 652        | 50.06        | 41.44        | +37.77        | 5        |

Bias% =  $100 \times \text{mean}((\text{model} - \text{market})/\text{market})$ .

## Page 4 · Analysis 1 · Overall moneyness · Tables M4-M5

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

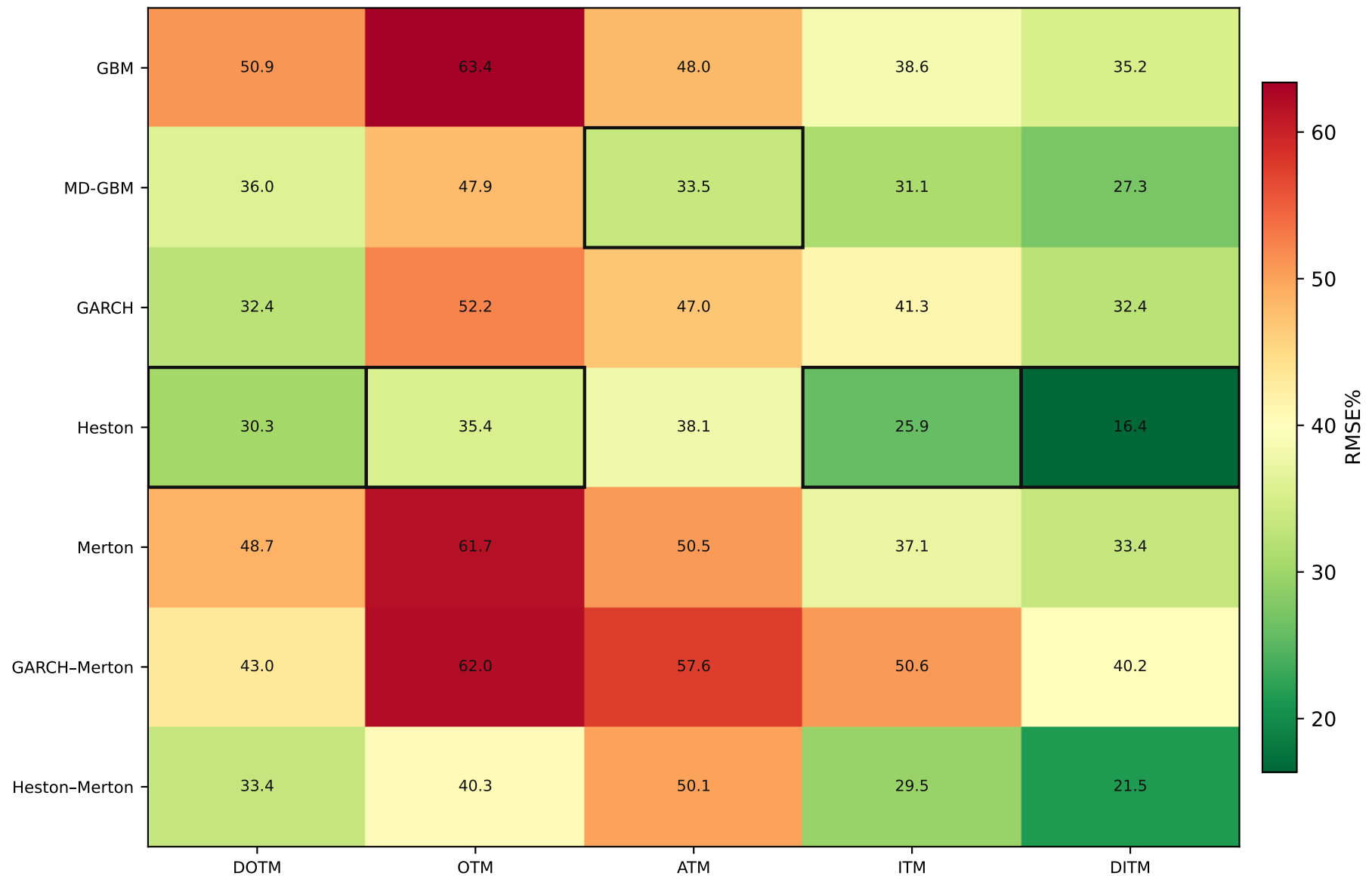
**Table M4 · ITM ·  $1.005 < m \leq 1.02$  · n = 58 · BEST = Heston**

| Model         | n         | RMSE%        | MAE%         | Bias%        | Ranking  |
|---------------|-----------|--------------|--------------|--------------|----------|
| GBM           | 58        | 38.61        | 31.48        | +13.60       | 5        |
| MD-GBM        | 58        | 31.08        | 25.55        | -2.12        | 3        |
| GARCH         | 58        | 41.32        | 32.10        | +24.66       | 6        |
| <b>Heston</b> | <b>58</b> | <b>25.88</b> | <b>21.53</b> | <b>+9.88</b> | <b>1</b> |
| Merton        | 58        | 37.15        | 30.51        | +19.81       | 4        |
| GARCH-Merton  | 58        | 50.60        | 40.78        | +37.89       | 7        |
| Heston-Merton | 58        | 29.55        | 25.77        | +16.08       | 2        |

**Table M5 · DITM ·  $m > 1.02$  · n = 16 · BEST = Heston**

| Model         | n         | RMSE%        | MAE%         | Bias%         | Ranking  |
|---------------|-----------|--------------|--------------|---------------|----------|
| GBM           | 16        | 35.18        | 28.19        | +16.72        | 6        |
| MD-GBM        | 16        | 27.26        | 23.27        | -0.08         | 3        |
| GARCH         | 16        | 32.43        | 24.65        | +19.14        | 4        |
| <b>Heston</b> | <b>16</b> | <b>16.35</b> | <b>14.20</b> | <b>+12.19</b> | <b>1</b> |
| Merton        | 16        | 33.37        | 24.52        | +23.76        | 5        |
| GARCH-Merton  | 16        | 40.15        | 31.54        | +31.54        | 7        |
| Heston-Merton | 16        | 21.49        | 19.00        | +17.43        | 2        |

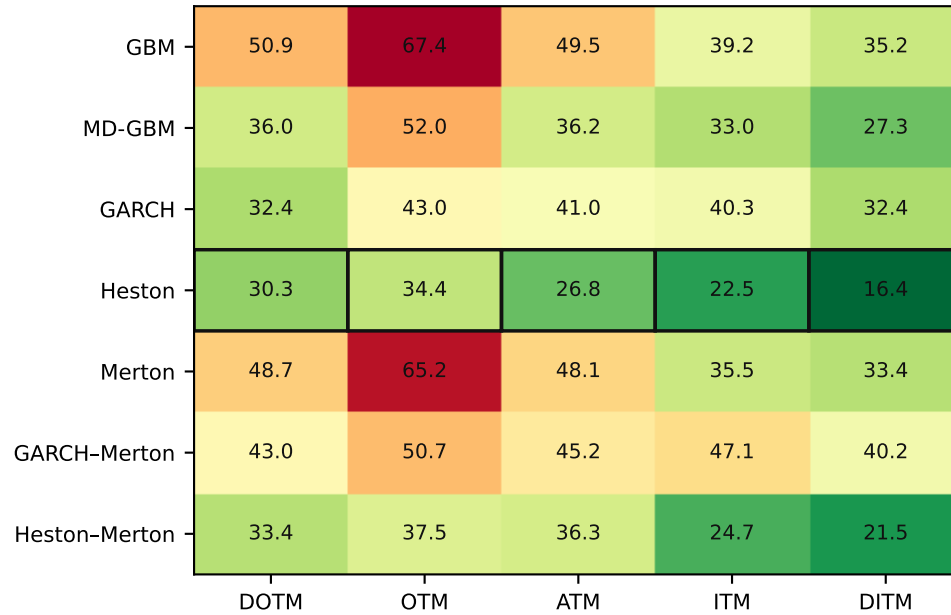
ITM/DITM:  $S > K$ . DOTM/OTM:  $S < K$ . ATM:  $|S/K - 1| \leq 0.5\%$ .



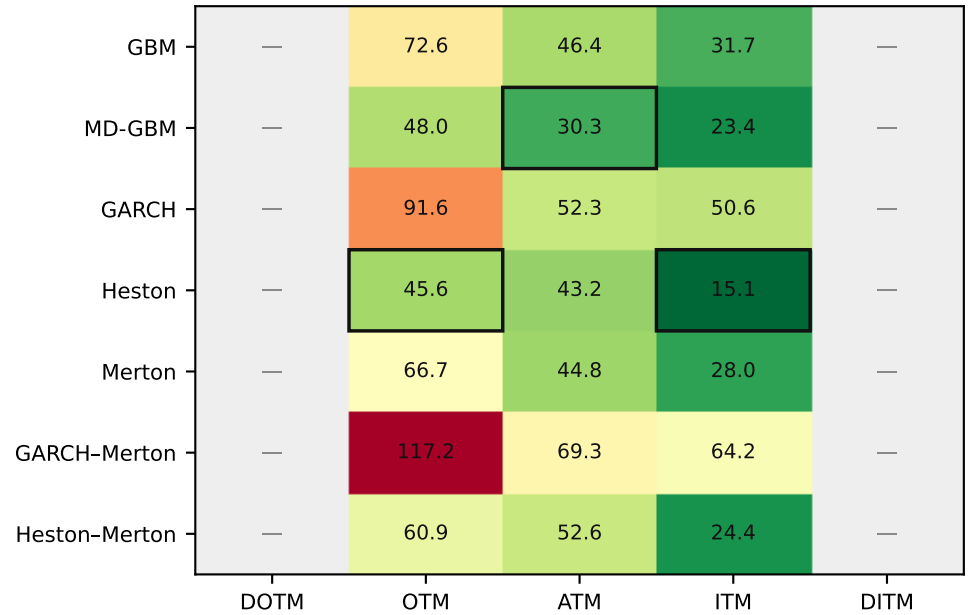
Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

**Figure M2 · RMSE% by model × moneyness · each window (companies pooled)**

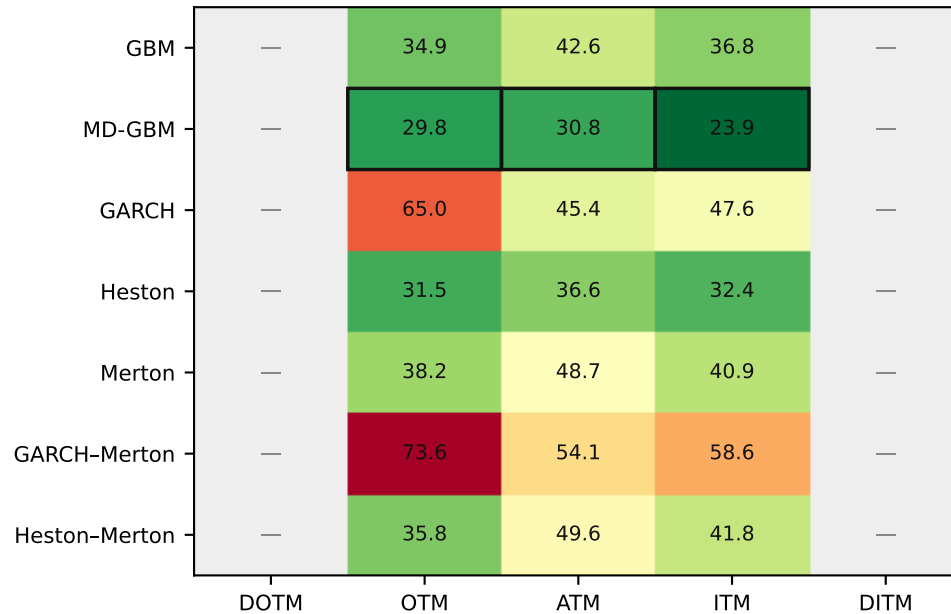
**2008-2009 · Crisis**



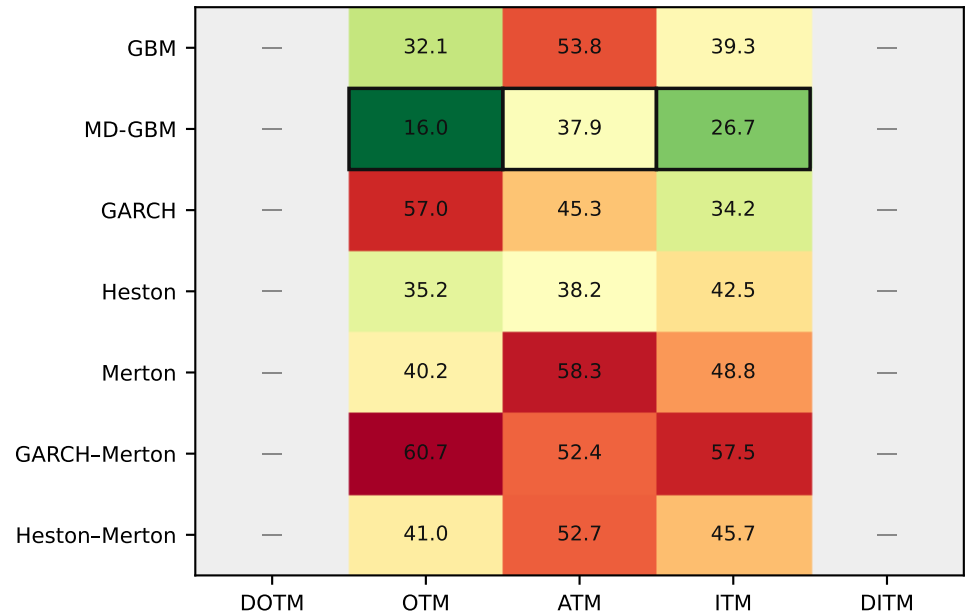
**2013-2014 · Normal**



**2018-2019 · Late-cycle**



**2019-2020 · COVID**



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

## Analysis 2 · Does the ranking change with moneyness inside each window?

**Table M6 · Best model (lowest RMSE%) by window × moneyness**

| Window               | DOTM   | OTM    | ATM    | ITM    | DITM   |
|----------------------|--------|--------|--------|--------|--------|
| 2008-2009 Crisis     | Heston | Heston | Heston | Heston | Heston |
| 2013-2014 Normal     | —      | Heston | MD-GBM | Heston | —      |
| 2018-2019 Late-cycle | —      | MD-GBM | MD-GBM | MD-GBM | —      |
| 2019-2020 COVID      | —      | MD-GBM | MD-GBM | MD-GBM | —      |

**Table M7 · n contracts by window × moneyness (shared across models)**

| Window               | DOTM | OTM | ATM | ITM | DITM |
|----------------------|------|-----|-----|-----|------|
| 2008-2009 Crisis     | 18   | 43  | 80  | 43  | 16   |
| 2013-2014 Normal     | 0    | 5   | 192 | 3   | 0    |
| 2018-2019 Late-cycle | 0    | 4   | 191 | 7   | 0    |
| 2019-2020 COVID      | 0    | 6   | 189 | 5   | 0    |

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that window. Columns with n = 0 cannot change the ranking.

### Does performance change with moneyness?

- DOTM (n = 18): BEST = Heston with RMSE% = 30.32, Bias% = +22.85.
- OTM (n = 58): BEST = Heston with RMSE% = 35.39, Bias% = +21.68.
- ATM (n = 652): BEST = MD-GBM with RMSE% = 33.54, Bias% = +10.33.
- ITM (n = 58): BEST = Heston with RMSE% = 25.88, Bias% = +9.88.
- DITM (n = 16): BEST = Heston with RMSE% = 16.35, Bias% = +12.19.
- Overall, the ranking does change with moneyness: DOTM → Heston, OTM → Heston, ATM → MD-GBM, ITM → Heston, DITM → Heston.
- Inside regimes, the BEST model also changes with moneyness in: 2013-2014.
- These comparisons use the Monday nearest-ATM sample, so ATM is the only well-populated bucket after 2009. DOTM/DITM results rest on few 2008–2009 quotes and should be read as descriptive, not as a high-powered test.